Javier Noe Ramos Silva

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Resume

Doctoral Researcher at UC Irvine specializing in quantum sensing and RF/MW engineering, with 7+ years leading large-scale radiocommunication projects for Mexican federal and military agencies. Track record managing teams in radar system development, mission-critical P25 networks, and microwave infrastructure design. M.Sc. in Telecommunications (IPN, Cum Laude) with publications in high-impact journals and international research collaboration (UAB, Spain).

Experience

Graduate Student Researcher September 2024 - Present

University of California, Irvine Irvine, USA

- Conducting doctoral research in quantum sensing and RF/MW engineering under Prof. Peter Burke, focusing on nitrogen-vacancy (NV) center-based biosensing applications.
- Teaching Assistant for EECS 170B (Electronics II) and EECS 195A (Drones), 150+ undergraduate students.
- Developing experimental protocols for single NV center detection and characterization using confocal microscopy, microwave control systems, and automated data acquisition/processing.

Radar System Development Coordinator January 2024 - September 2024

Secretaría de la Defensa Nacional (SEDENA) Mexico City

- Technical Lead of a 7+ engineer antenna engineering team for the "Research and Development of a 3D Air Surveillance Radar System TPS-CAMAZOT-220", overseeing antenna design, RF validation, and system-level performance characterization developed for national military airspace surveillance operations.

Engineering Manager (Microwave, Optical Fiber and Data Analysis) July 2023 - April 2024

Gobierno del Estado de Sonora Mexico City

- Led engineering team (5+ members) in designing state-level telecommunications trunk network, coordinating fiber optic backbone, microwave link planning, and infrastructure analysis for the "Design of the Trunk Network of the State of Sonora" project.
- Conducted comprehensive diagnostics of Sonora's telecommunications infrastructure, analyzing network coverage, capacity constraints, and modernization requirements.
- Developed data collection and analysis tools with interactive web-based mapping platform (JavaScript, HTML5, CSS3, QGIS) for infrastructure visualization and project reporting.

Microwave Network Design Specialist Consultant October 2022 - July 2023

Corredor Interoceánico del Istmo de Tehuantepec (CIIT) Mexico City

- Designed microwave backbone network for interoceanic corridor, analyzing service requirements and developing technical/economic proposals integrating PtP/PMP microwave links and fiber optic segments.
- Conducted network planning and coverage analysis using MLinkPlanner and QGIS, sizing microwave links across the corridor spanning Pacific to Atlantic coasts.
- Evaluated market solutions for network implementation and delivered technical documentation, executive presentations, and web-based project tracking tools.

RF/MW R&D Engineer/Scientist October 2020 - February 2022

SEMAR UNINDETEC/INIDETAM Veracruz

- Led RF/MW engineering for "Development of Mexican Radiocommunications P25 (Vírgula Project)" (FORDECYT-funded), designing and implementing mission-critical P25 radiocommunication systems for Mexican Navy operations.
- Designed, simulated, and characterized RF circuits including power amplifiers, LNAs, mixers, filters, and transceivers using ADS Keysight, performing system-level integration and board level validation.
- Developed firmware (C, VHDL) for microcontroller and FPGA-based systems with SPI interfaces, conducting laboratory characterization using VNA, spectrum analyzers, and RF test equipment.
- Trained military and civilian personnel in RF engineering, digital/analog electronics, and antenna systems.

MW Customer Support Engineer March 2020 - October 2020

Huawei Technologies de México Querétaro

– Provided technical support for Huawei microwave transmission systems, analyzing incidents and troubleshooting performance issues for telecommunications carrier clients.

– Conducted link performance analysis and optimization, interfacing with field engineers and network operations teams.

Mission Critical Radio Specialist Consultant January 2018 - August 2018

Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública Mexico City

– Developed "Prospective Study on the National Radiocommunication Network" defining strategic modernization plan for Mexico's public safety communications infrastructure, analyzing system architecture, migration scenarios, and short/medium/long-term roadmap.

– Completed intensive vendor training in mission-critical radio technologies (P25, Tetrapol, TETRA, LTE) from Motorola, Airbus, and Huawei, informing technical evaluation and strategic recommendations for network upgrade.

Antenna Design and Calibration Engineer June 2017 - July 2018

Laboratorio Nacional en Telecomunicaciones y Antenas (LaNTA-IPN) Mexico City

– Designed and characterized antennas using EM simulation tools (ADS, CST, Altair Feko) and laboratory equipment (VNA, anechoic chamber, spectrum analyzers), conducting electromagnetic propagation analysis and field measurements.

– Developed measurement methodologies for technical validation and supported laboratory accreditation procedures for EMA, ISO 9001:2015 and ISO/IEC 17025:2005.

Education

2024 - 2028 Ph.D. in Electrical and Computer Engineering
University of California, Irvine Irvine, USA

2018 - 2020 Master of Science in Telecommunications Engineering (Cum Laude)
Instituto Politécnico Nacional Mexico City

2019 - 2020 Research Stay (Fully funded)
Universitat Autònoma de Barcelona Barcelona, Spain

2014 - 2017 Bachelor of Communications and Electronics Engineering
Instituto Politécnico Nacional Mexico City

Projects

Investigación y Desarrollo de un Sistema de Radar de Vigilancia Aérea 3D TPS-CAMAZOT-220 Link

Design and implementation of a radar system for the Mexican Army.

Diseño de la Red Troncal del Estado de Sonora Link

Design of the trunked network of Sonora, México.

Corredor Interoceánico de Telecomunicaciones y Servicios Digitales Link

Development of the telecommunications and digital services platform of the Corredor Interoceánico del Istmo de Tehuantepec (CIIT).

Desarrollo de Radiocomunicaciones Mexicanas P25 (Proyecto Vírgula) Link

Design and implementation of mission-critical P25 radio communication systems for the Mexican Navy.

Estudio prospectivo sobre la Red Nacional de Radiocomunicación Link

Prospective study on the National Radiocommunication Network, which contemplates the short, medium and long-term strategic plan, which aims to guarantee the operation, maintenance and updating of the network, in accordance with the existing technology in the market, in compliance with the agreement 10/XLII/17 of the National Public Security Council.

Skills

Languages	Spanish (Native), English (Advanced - TOEFL iBT 97, CEFR C1).
RF/MW	ADS, CST, Feko, MLinkPlanner, VNA, Spectrum Analyzer, RF Generator, Anechoic Chamber, PIM.
Programming	C, C++, Python, MATLAB, VHDL, Verilog-A, JavaScript, HTML5, CSS3, NodeJS, MySQL, MongoDB.
Cloud & Tools	Google Cloud Platform (Certified Associate Engineer), LaTeX, Linux, QGIS.

Awards & Honors

Fellowship	Henry Samueli Endowed Fellowship (2026)
Fellowship	UCI Electrical Engineering and Computer Science Department (2024-2025)
Scholarship	Mexico's National Council of Science and Technology M.Sc. (2018-2020).
Scholarship	Mexico's National Council of Science and Technology International Ph.D. (2023-2027), declined.
Grant	International Academic Mobility Grant (IPN, 2019)
Grant	Institutional Research Training Program (PIFI) (2015-2018).
Honorable mention	Fourth science and technology contest "Vive conCiencia 2017" (National Finalist)

Peer Review

Peer Reviewer	Springer Nature (Discover Applied Sciences), IOP Publishing (Nanotechnology, Physica Scripta, Engineering Research Express, Semiconductor Science and Technology), AEU (International Journal of Electronics and Communications).

Certificates

Google	Associate Cloud Engineer Certification
Google	Google Data Analytics Professional Certificate
Huawei	HCCDA-Tech Essentials (Huawei Cloud) ID: HWDCTEDA402783
IOP Publishing	2023 Online Peer Review Excellence Graduates
Udemy	100 Days Of Code - 2023 Web Development Bootcamp
Mentor Collective	Student Success Mentor Training

Teaching

EECS 170B Electronics II (LAB)	TA UC Irvine
EECS 195A Drones (LAB)	TA UC Irvine
Electromagnetic Radiator Theory	TA ESIME ZAC
Microwave Circuits	TA ESIME ZAC

Extracurricular

Housing Assitant	GFH (Graduate & Family Housing) (UCI) (2025-Present)
Member/Treasurer	MGREP (Mexico Graduate Research & Education Program) (UCI) (2023-Present)
Volunteer Teacher	Use of LaTeX for Scientific Writing, Engineering and Programming (2016-2020)
Volunteer	Nursing Home for the Elderly "Los Tamayo" (Oaxaca) (2016-2018)
Athlete	Competitive Team Athletics (10 km/ Half-Marathon) & Triathlon at IPN (2013-2016)

Publications

Noh, Sangjun et al. (2026). "Highly Localized Delivery of Microwaves to Quantum Bits with a Scanning Loop Micro-Antenna". In: *2026 IEEE MTT-S Radio Frequency Systems and Applications Symposium (IMS RFSA)*, pp. 337–340. DOI: 10.1109/IMSRFSA70221.2026.11624215.

Pacheco-Sanchez, Anibal, J. Noé Ramos-Silva, et al. (2024). "High-Data-Rate Modulators Based on Graphene Transistors: Device Circuit Co-Design Proposals". In: *Electronics* 13.20. ISSN: 2079-9292. DOI: 10.3390/electronics13204022. URL: https://www.mdpi.com/2079-9292/13/20/4022.

Pacheco Sanchez, Anibal et al. (2023). "Design of GFET-based active modulators leveraging device performance reproducibility conditions". In: *2023 38th Conference on Design of Circuits and Integrated Systems (DCIS)*, pp. 1–6. DOI: 10.1109/DCIS58620.2023.10335972.

Pasadas, Francisco et al. (2023). "Exploiting Ambipolarity in Graphene Field-Effect Transistors for Novel Designs on High-Frequency Analog Electronics". In: *Small* 19.49, p. 2303595. DOI: https://doi.org/10.1002/smll.202303595. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/smll.202303595. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/smll.202303595.

Pacheco–Sanchez, Anibal et al. (2022). "Exploiting the ambipolarity in emerging transistors for high-frequency applications". In: *2022 37th Conference on Design of Circuits and Integrated Circuits (DCIS)*, pp. 01–06. DOI: 10.1109/DCIS55711.2022.9970151.

Pacheco-Sanchez, Anibal, Javier N. Ramos-Silva, et al. (2021). "A Small-Signal GFET Equivalent Circuit Considering an Explicit Contribution of Contact Resistances". In: *IEEE Microwave and Wireless Components Letters* 31.1, pp. 29–32. DOI: 10.1109/LMWC.2020.3036845.

Ramos-Silva, Javier N., Aníbal Pacheco-Sanchez, Eloy Ramirez-Garcia, et al. (2021). "Towards the development of multifunctional circuits using emerging carbon-based devices". In: *UAB Divulga*.

Ramos-Silva, Javier N., Aníbal Pacheco-Sánchez, et al. (2021). "Multifunctional High-Frequency Circuit Capabilities of Ambipolar Carbon Nanotube FETs". In: *IEEE Transactions on Nanotechnology* 20, pp. 474–480. DOI: 10.1109/TNANO.2021.3082867.

Ramos-Silva, Javier N et al. (2020). "Small-signal parameters extraction and noise analysis of CNTFETs". In: *Semiconductor Science and Technology* 35, p. 045024. DOI: 10.1088/1361-6641/ab760b.

Ramos-Silva, Javier N., Aníbal Pacheco-Sanchez, Luis M. Díaz-Albarran, et al. (2020). "High-Frequency Performance Study of CNTFET-Based Amplifiers". In: *IEEE Transactions on Nanotechnology* 19, pp. 284–291. DOI: 10.1109/TNANO.2020.2978816.

Ramos-Silva, Javier N., Eloy Ramírez-García, et al. (2019). "Design of a Compact Ultra Wide Band Flexible Antenna for Personal Mobile Communications". In: *2019 IEEE International Fall Meeting on Communications and Computing (ROC&C)*. DOI: 10.1109/ROCC.2019.8873531.